

KONSTANTINOS ZERVAKIS

AI Engineer — LLM Applications, RAG & Agentic Systems

Heraklion, Crete, Greece — open to fully remote (EU / UK / US time zones)

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Summary

AI engineer building production systems with large language models: Retrieval-Augmented Generation (RAG), multi-agent orchestration, and the evaluation and observability infrastructure that keeps non-deterministic output reliable. Five systems shipped across industrial automation, agriculture, aerospace verification, scientific benchmarking and education. Currently owns automated verification tooling at OHB Hellas under ECSS V&V/PA standards; PhD candidate in reinforcement learning for autonomous navigation. EU work authorisation. Greek (native), English (C2).

Technical Skills

LLM APIs & SDKs	Anthropic Claude API, OpenAI API, Anthropic Agent SDK, Model Context Protocol (MCP), FastMCP, function calling, structured output, system prompts, guardrails
Agents & orchestration	Pydantic AI, pydantic-graph, LangChain, LlamaIndex, multi-agent systems, DAG decomposition, ReAct, Reflexion, typed dependency injection, output validators with retries
RAG & vector search	Retrieval-Augmented Generation (RAG), FAISS, ChromaDB, vector databases, hybrid search (BM25 + dense), BGE-M3 embeddings, cross-encoder reranking, semantic search, fine-tuning, LoRA
Evaluation & observability	Golden-set oracles, regression testing of non-deterministic output, promptfoo, Langfuse, OpenTelemetry (GenAI), prompt versioning and registries, hallucination / grounding checks, pytest, MLflow
Infrastructure	Python (async, typed), FastAPI, Docker, Kubernetes, GitHub Actions, CI/CD, PostgreSQL, Redis, Ray, Spark, Prometheus, Grafana, uv
Machine learning	PyTorch, scikit-learn, LightGBM, ONNX, reinforcement learning, time-series forecasting
Languages	Python, SQL, C, C++ , Bash, MATLAB, XML

Experience

Programme Leader & AI Faculty — MBS College of Crete from Oct 2026
Heraklion, Greece

- Appointed Programme Leader; teaching applied artificial intelligence and machine learning at postgraduate (MSc) level.

Software Test Engineer — OHB Hellas (Aerospace) Oct 2025 – present
Crete, Greece

- Built automated test-generation pipelines in Python for satellite ground-segment software, reaching 100% requirement-to-test traceability and cutting verification cycle time by an estimated 3× against the previous manual workflow.
- Delivered quality documentation aligned to ECSS V&V/PA standards that passed external review with zero rework.
- Authored the team's standard testing reference guide, now the onboarding baseline for junior engineers.
- Represented OHB Hellas at InnoDays 2025, the Region of Crete innovation event, Heraklion.

Selected Projects

AIRTH Vinea — Production LLM Agent System for Agronomic Decisions github.com/zervakisai/vinea
Python, Pydantic AI, PostgreSQL, FastAPI, Docker, OpenTelemetry, Langfuse

- Built a multi-agent LLM system (3 agents on a typed graph) that turns hourly weather data into daily irrigation and spray advisories for vineyards, with structured outputs and validator-enforced grounding — the model cannot emit a number it did not receive.
- Designed the LLM/deterministic boundary as the core architecture: all physics (FAO-56 water balance, spray-window detection) runs in plain Python; the LLM is a bounded judgement-and-explanation layer. The core is verifiably unchanged across all 8 production phases (git diff on core modules: empty).
- Shipped the full production stack in 12 tagged phases: PostgreSQL persistence with provenance tracking (model, prompt version, config hash per advisory), a Postgres-backed work queue with SKIP LOCKED for concurrent batch workers, distributed tracing (OpenTelemetry → self-hosted Langfuse), a FastAPI service, and a Streamlit operations dashboard.
- Built an LLM evaluation pipeline gated in CI, scored by deterministic ground-truth oracles instead of an LLM judge, plus a prompt registry with name@label versioning and eval-gated promotion — prompts change without redeploying.

- Documented every architectural call in 5 ADRs with rejected alternatives (e.g. Postgres queue over Redis: no second stateful system until it earns its place).

Additional engineering work

- Predictive failure prevention from live telemetry — deployed an LSTM / LightGBM / ETSformer ensemble over six real-time sensor streams at sub-5 ms inference on embedded hardware, wired into a pipeline of ingestion, inference, threshold alerting and automated intervention.
- Large-scale document processing — extracted values from 14M+ documents at under 2% error inside a 20 KB memory budget using Spark.
- Sensor-based classification (HAR-Drive) — 98.4% F1 with 98% feature reduction, holding accuracy on constrained production-line hardware.

Education

PhD candidate — Autonomous navigation of UAVs. Hellenic Army Academy. Dec 2025 – present
Supervisor: Assoc. Prof. Ilias Panagiotopoulos. Reinforcement learning and quantum machine learning.

MSc Machine Learning and Data Science. Technical University of Crete. Sep 2025

BSc Mathematics and Applied Mathematics. University of Crete, 241 ECTS. Jul 2024

Research and Publications

- **FLARE: decision optimization under uncertainty** — lead author, submitted to *Sensors* (MDPI), 2026. github.com/zervakisai/flare_n. 450+ experimental runs showing safety-tuned planners raise total cost by 23–38% under timing constraints; introduces a configurable risk coefficient ρ . Friedman $\chi^2(4) = 190.8$ ($p < 0.001$), Wilcoxon signed-rank with Bonferroni correction, Cliff's δ .
- **AI for personalized learning in secondary education** — peer-reviewed, 2026, with Charalampos Zervakis (M.Ed.). Live-deployed adaptive learning platform with automated content generation and structured accessibility adaptations; pilot measured gains in engagement, comprehension and teacher productivity.
- **Hybrid reinforcement learning and quantum machine learning** — doctoral research, Hellenic Army Academy. Variational quantum circuits combined with RL (PyTorch, PennyLane) for autonomous multi-agent coordination. Two papers in preparation (FLARE-PO, FLARE-Q) targeting NeurIPS/ICRA and *npj Quantum Information*.

Speaking and Distinctions

Live radio interview on AI and autonomous systems, SKAI Crete FM | long-form career podcast on agentic AI and space technology | technical essay on AI orchestration and the risk coefficient ρ (Medium) | two-time silver medalist in fencing, national and world championships.